Single-Sided and Fused Selected-Space CI Transport Contractions

Theory and implementation note

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1 Selected space and required derivatives

Let W denote the fixed expansion of the selected VB structures into unique spin-string products. The orthogonal-carrier image of a selected coefficient vector c is

$$F(R)c = \mathrm{vec}\left[T_\alpha(R) X_c T_\beta(R)^\mathsf{T}\right], \qquad \mathrm{vec}(X_c) = Wc, \tag{1}$$

where $T_\sigma(R) = \wedge^{n_\sigma} R$, and R is the upper-triangular active orbital factor. The matrix X_c is a mathematical representation only: the implementation reads its touched columns through the sparse structure map. Vectorization uses the fixed product index $aU_\beta+b$, where a and b label alpha and beta strings, respectively. No selected-space closure under $T_\alpha \otimes T_\beta$ is assumed.

For two independent orbital factors, define

$$f(L, R; \lambda, c) = \lambda^\mathsf{T} F(L)^\mathsf{T} F(R) c. \tag{2}$$

The left and right cotangents are independent derivatives of this scalar. A caller requesting only one must not compute the other. The identity

$$f(L, R; \lambda, c) = f(R, L; c, \lambda) \tag{3}$$

allows a left-only contraction to use the right-only kernel with both charts, coefficients and supplied chart directions exchanged. It holds for any real selected structure map W, including a nonclosed subspace.

At an accepted point $L = R = R_0$, the transport curvature consumes three objects:

$$G_d = D_R \nabla_R f(R_0, R_0; \lambda, c)[\dot{R}], \tag{4}$$

$$G_q = \nabla_R f(R_0, R_0; \lambda, q), \tag{5}$$

$$t = DF(R_0)[\dot{R}]^\mathsf{T} F(R_0)\lambda. \tag{6}$$

Here λ is the accepted residual inverse image, c the accepted CI vector, q its independent direction, and $\dot{R}$ the orbital direction. The left chart is held fixed in G_d. None of these terms is discarded or approximated. The mixed-block reciprocity is

$$\langle G_q, \dot{R} \rangle = q^\mathsf{T} t. \tag{7}$$

2 Common tile intermediates

All exterior matrices and directions below are evaluated at R_0. For a tile J of original beta strings, let E_J select their unit vectors. Set

$$U = T_\alpha X_c E_J, \qquad V = T_\beta E_J, \qquad U_q = T_\alpha X_q E_J, \tag{8}$$

$$A = T_\alpha^\mathsf{T} U, \qquad B = T_\beta^\mathsf{T} V, \qquad A_q = T_\alpha^\mathsf{T} U_q, \tag{9}$$

$$Z_\alpha = T_\alpha X_\lambda B, \qquad Z_\beta = T_\beta X_\lambda^\mathsf{T} A, \qquad Z_{\beta q} = T_\beta X_\lambda^\mathsf{T} A_q. \tag{10}$$

Products with X_λ use only its original touched products. The same V, B and Z_α serve the accepted and independent-CI directions. Since only the right chart moves,

$$\dot{U} = \dot{T}_\alpha X_c E_J, \qquad \dot{V} = \dot{T}_\beta E_J, \tag{11}$$

$$\dot{A} = T_\alpha^\top \dot{U}, \qquad \dot{B} = T_\beta^\top \dot{V}, \tag{12}$$

$$\dot{Z}_\alpha = T_\alpha X_\lambda \dot{B}, \qquad \dot{Z}_\beta = T_\beta X_\lambda^\top \dot{A}. \tag{13}$$

The fixed left exterior factors are not differentiated in these expressions.

Define $\mathcal{P}_\sigma(Z, Y)$ as the upper-chart cotangent of $\langle Z, T_\sigma C\rangle$ with respect to its right factor, where $Y = T_\sigma C$. Its directional implementation differentiates the seed, image and chart pullback, including the inverse-factor derivative already present in the exterior generator pullback. The required contractions are

$$G_d = \sum_J D\mathcal{P}_\alpha(Z_\alpha, U)[\dot{Z}_\alpha, \dot{U}, \dot{R}] + D\mathcal{P}_\beta(Z_\beta, V)[\dot{Z}_\beta, \dot{V}, \dot{R}], \tag{14}$$

$$G_q = \sum_J \mathcal{P}_\alpha(Z_\alpha, U_q) + \mathcal{P}_\beta(Z_{\beta q}, V). \tag{15}$$

Only these right cotangents are evaluated; their unused left counterparts do not belong to either result.

3 Eliminating the separate transpose-action sweep

The raw coefficient-space tile of the transpose directional image is

$$(t_{\mathrm{raw}})_J = \dot{T}_\alpha^\top Z_\alpha + T_\alpha^\top \dot{Z}_\alpha. \tag{16}$$

Both seeds on the right-hand side have already been obtained for G_d. An exterior directional-adjoint action therefore produces this tile without another expansion of λ or an independent overlap-direction sweep. For clarity, expanding the seeds gives

$$(t_{\mathrm{raw}})_J = \dot{T}_\alpha^\top T_\alpha X_\lambda T_\beta^\top T_\beta E_J \tag{17}$$

$$+ T_\alpha^\top T_\alpha X_\lambda T_\beta^\top \dot{T}_\beta E_J. \tag{18}$$

These are precisely the alpha and beta contributions to $DF(R_0)[\dot{R}]^\top F(R_0)\lambda$. The final result is

$$t = W^\top \operatorname{vec}(t_{\mathrm{raw}}). \tag{19}$$

Only entries touched by W are sampled from each tile. This is an exact projection, not a completion of the selected space. Equation (7) also follows immediately and provides an independent transpose check.

4 Memory and unchanged numerical model

At most eight alpha and seven beta row panels coexist. For width b their payload is bounded by $8b(8U_\alpha + 7U_\beta)$ bytes. The planner uses the conservative bound $64b(U_\alpha + U_\beta)$ within one 8 MiB panel allowance, or the unavoidable one-column working set. One shared touched-product vector contains $W\lambda$; Wc and Wq are generated only for the current beta tile. The forcing is accumulated directly into the selected coefficients. This factor order has no transformed spin-product image, pair-squared Gram, or worker-owned replica of these arrays. Workers own disjoint row ranges of shared panels.

When the touched support is the complete product carrier, the existing one-body-moment order can be less expensive than the factor order. Its fused implementation retains three shared product images only if they fit the existing 64 MiB overlap allowance, not a larger workspace. These are $L = F\lambda$, $Y = Fc$, and $\dot{Y} = DF[\dot{R}]c$. The right cotangent and its direction use the one-body moments of (L, Y) and $(L, \dot{Y})$. After this contraction, the storage of Y is reused for Fq and released. The dead $\dot{Y}$ storage then holds the

transpose directional image obtained by applying the directional exterior adjoints to L. Thus the broad-support order also shares its accepted image and avoids an additional overlap sweep. If these three images do not fit, the same exact operator uses the bounded factor sweep. This selects storage and contraction order, not the physical state space; neither selected-space closure nor an approximation is invoked.

The full physical transport curvature, selected metric solves, orbital gradient/HVP pullbacks and outer convergence criteria remain unchanged. No residual threshold, rank cutoff, amplitude cutoff or CI-state approximation is introduced. Tests compare the fused contraction with independently evaluated cross-Gram derivatives, explicit occupied-minor finite differences, and Eq. (7), including unequal spins, restricted supports, nonclosed selections, zero directions and singular occupied minors.

5 Finite-keyframe pullback in one sweep

Let R and R_0 denote the current and reference charts, respectively, and let $\dot{R}_0$ be the accumulated reference direction. For a single state, write

$$y_\lambda = F(R)\lambda, \quad y_c = F(R)c, \quad y_0 = F(R_0)c_0, \quad z = F(R_0)b + DF(R_0)[\dot{R}_0]c_0 - y_c. \tag{20}$$

Here b is the reference coefficient plus its independent CI displacement; c is the projected transported coefficient, not an assumption that the target lies in the selected space. With λ and these coefficients held fixed, the two cotangents consumed by the finite transport are

$$\langle G_R, \delta R\rangle = \langle DF(R)[\delta R]\lambda, z\rangle - \langle y_\lambda, DF(R)[\delta R]c\rangle, \tag{21}$$

$$\langle G_0, \delta R_0\rangle = \langle y_\lambda, DF(R_0)[\delta R_0]c_0\rangle. \tag{22}$$

The first line contains *both* sides of the current metric derivative. The second is the cotangent of the reference direction used in the transport map; it does not differentiate the reference chart of the entire first line. The CI-gradient consumer additionally requires

$$t = F(R_0)^{\mathsf{T}} y_\lambda. \tag{23}$$

For an original beta tile J, form

$$U_0 = T_{0\alpha} X_{c_0} E_J, \qquad V_0 = T_{0\beta} E_J, \tag{24}$$

$$\dot{U}_0 = \dot{T}_{0\alpha} X_{c_0} E_J, \qquad \dot{V}_0 = \dot{T}_{0\beta} E_J, \tag{25}$$

$$U_b = T_{0\alpha} X_b E_J + \dot{U}_0, \qquad U_c = T_\alpha X_c E_J, \qquad V_c = T_\beta E_J. \tag{26}$$

The target is the sum of $U_b V_0^{\mathsf{T}}$, $U_0 \dot{V}_0^{\mathsf{T}}$, and $-U_c V_c^{\mathsf{T}}$. For any of these pairs (U, V), common current-chart seeds are

$$Z_\alpha = T_\alpha X_\lambda T_\beta^{\mathsf{T}} V, \qquad Z_\beta = T_\beta X_\lambda^{\mathsf{T}} T_\alpha^{\mathsf{T}} U. \tag{27}$$

The first pair's Z_α and the second pair's Z_β also supply G_0. Applying $T_{0\alpha}^{\mathsf{T}}$ to that first Z_α and sampling only the original touched entries supplies t. Thus the reference cotangent and CI image need no independent cross-Gram sweep.

The factor planner conservatively allows sixteen alpha-plus-beta panels per tile within the existing 64 MiB overlap allowance. There is one shared $W\lambda$ vector, no pair-squared Gram or worker-owned panel replicas. Complete broad support retains a moment contraction when five shared product images fit that same allowance; otherwise the identical operator is streamed.

6 Fixed-state energy and adjoint sharing

For m equally weighted prescribed states, define

$$G = C^{\mathsf{T}} M C, \qquad K = C^{\mathsf{T}} H C, \qquad E = w \operatorname{Tr}(G^{-1} K), \qquad w = 1/m. \tag{28}$$

Only MC is needed before normalization. Factor $G = LL^{\mathsf{T}}$, set $V = L^{-\mathsf{T}}$, and let $\widetilde{C} = CV$. A single shared sigma/adjoint sweep produces $H\widetilde{C}$ and the unshifted integral cotangents. With $\widetilde{K} = \widetilde{C}^{\mathsf{T}} H\widetilde{C}$, the fixed-coefficient variation is

$$\delta E = w \operatorname{Tr}(\widetilde{C}^{\mathsf{T}} \delta H \widetilde{C}) - w \operatorname{Tr}(\widetilde{K}\widetilde{C}^{\mathsf{T}} \delta M \widetilde{C}). \tag{29}$$

The complete multiplier $-\widetilde{K}$, including its off-diagonal entries, therefore completes the overlap cotangent after the same sweep. No second Hamiltonian action is necessary. A small state-space Ritz transformation may subsequently provide individual energies and representatives; equal-weight energy and orbital cotangents are invariant under that transformation. The derivative in the original coefficient frame is

$$\nabla_C E = 2w\big(H\widetilde{C} - M\widetilde{C}\widetilde{K}\big)V^{\mathsf{T}}. \tag{30}$$

All matrices inverted or diagonalized in this normalization are m by m, not full structure-space matrices. The implementation is in `assembly/states.*`; bounded finite-transport contractions are in `assembly/transport_tiles.cpp`. Neither operation changes the selected VB space, its spin carrier, the optimizer strategy, or its accuracy criteria.